

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 5: Arithmetic Progressions

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sum of the first n terms

1. The sum of the first m terms of an AP is $2m^2 + 3m$. Find its second term. **CBSE 2025, 1 mark**

SOLUTION

$$S_1 = 2 + 3 = 5, \text{ so the first term is } 5.$$

$$S_2 = 2(4) + 3(2) = 14$$

$$\text{Second term} = S_2 - S_1 = 14 - 5$$

Answer: 9

2. An A.P. has first term $a = 7$ and n th term $a_n = 84$, and its first n terms add up to $\frac{2093}{2}$. Find n . (a) 22 (b) 24 (c) 23 (d) 26 **CBSE 2024, 1 mark**

SOLUTION

$$S_n = \frac{n}{2}(a + a_n) = \frac{n}{2}(7 + 84) = \frac{91n}{2}$$

$$\frac{91n}{2} = \frac{2093}{2} \text{ gives } n = 23$$

Answer: (c)

3. In an A.P., the first and eighth terms have sum 32 and product 60. Find the first term and the common difference, and then the sum of the first 20 terms. **CBSE 2024, 5 marks**

SOLUTION

$$a + (a + 7d) = 32, \text{ so } a + 7d = 32 - a.$$

$$\text{Product: } a(32 - a) = 60, \text{ so } a^2 - 32a + 60 = 0 \text{ and } (a - 2)(a - 30) = 0. \text{ Hence } a = 2 \text{ or } a = 30.$$

$$\text{If } a = 2: 7d = 28, \text{ so } d = 4. \text{ Then } S_{20} = \frac{20}{2}[2(2) + 19(4)] = 10 \times 80 = 800.$$

$$\text{If } a = 30: 7d = -28, \text{ so } d = -4. \text{ Then } S_{20} = \frac{20}{2}[2(30) + 19(-4)] = 10 \times (-16) = -160.$$

Answer: $a = 2, d = 4, S_{20} = 800$; or $a = 30, d = -4, S_{20} = -160$.

4. An A.P. has 40 terms. Its first 9 terms add up to 153 and its last 6 terms add up to 687. Find the first term, the common difference and the sum of all 40 terms. **CBSE 2024, 5 marks**

SOLUTION

$$S_9 = \frac{9}{2}[2a + 8d] = 153, \text{ so } a + 4d = 17.$$

The last 6 terms are a_{35} to a_{40} . Their sum $= \frac{6}{2}(a_{35} + a_{40}) = 3(2a + 73d) = 687$, so $2a + 73d = 229$.

Substituting $a = 17 - 4d$: $34 - 8d + 73d = 229$, so $65d = 195$ and $d = 3$. Then $a = 5$.

$$S_{40} = \frac{40}{2}[2(5) + 39(3)] = 20 \times 127 = 2540$$

Answer: $a = 5$, $d = 3$, and the sum of all 40 terms is 2540.

2 Case study

5. In a fund-raising run, the first round is 300 m long and every round after it is 50 m longer than the one before, so the rounds are 300 m, 350 m, 400 m and so on. There are 10 rounds in all. (i) Write the 4th, 5th and 6th terms of this AP. (ii) How long is the 8th round? (iii) Find the total distance covered in all 10 rounds, OR the total distance covered by a runner who stops after 6 rounds.

CBSE 2025, 4 marks

SOLUTION

The lengths form an AP with $a = 300$ and $d = 50$.

(i) $a_4 = 300 + 3(50) = 450$, $a_5 = 500$, $a_6 = 550$. **The terms are 450, 500 and 550.**

(ii) $a_8 = 300 + 7(50) = 650$. **The 8th round is 650 m.**

(iii) $S_{10} = \frac{10}{2}[2(300) + 9(50)] = 5 \times 1050 = 5250$. **Total for 10 rounds: 5250 m.**

OR: $S_6 = \frac{6}{2}[2(300) + 5(50)] = 3 \times 850 = 2550$. **Total for 6 rounds: 2550 m.**

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.